

Q1

AECBD, CBCAD

Q2

1. $\epsilon/6\epsilon_0 = 3 \times 10^{-9} \text{ V}\cdot\text{m}$ **3 points**

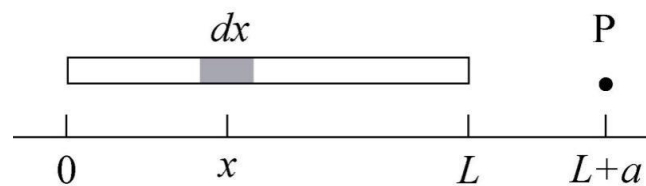
2. $2\pi AR^3/\mu_0$ **3 points**

Q3 10 points

(a) The linear charge density is the charge per unit length of rod. Since the charge is uniformly distributed on the rod,

$$\lambda = \frac{-q}{L} = \frac{-4.23 \times 10^{-15} \text{ C}}{0.0815 \text{ m}} = -5.19 \times 10^{-14} \text{ C/m}.$$

(b) We position the x axis along the rod with the origin at the left end of the rod, as shown in the diagram.



Let dx be an infinitesimal length of rod at x . The charge in this segment is $dq = \lambda dx$.

The charge dq may be considered to be a point charge. The electric field it produces at point P has only an x component, and this component is given by

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(L+a-x)^2}$$

The total electric field produced at P by the whole rod is the integral

$$\begin{aligned} E_x &= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(L+a-x)^2} = \frac{\lambda}{4\pi\epsilon_0} \left. \frac{1}{L+a-x} \right|_0^L = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{L+a} \right) \\ &= \frac{\lambda}{4\pi\epsilon_0} \frac{L}{a(L+a)} = -\frac{1}{4\pi\epsilon_0} \frac{q}{a(L+a)}, \end{aligned}$$

upon substituting $-q = \lambda L$. With $q = 4.23 \times 10^{-15} \text{ C}$, $L = 0.0815 \text{ m}$ and $a = 0.060 \text{ m}$, we obtain $E_x = -4.48 \times 10^{-3} \text{ N/C}$, or $|E_x| = 4.48 \times 10^{-3} \text{ N/C}$. The negative sign in E_x indicates that the field points in the $-x$ direction, or 180° counterclockwise from the $+x$ axis.

Q4 10 points

$$V = \frac{Q}{4\pi\epsilon_0 R} + \left(\frac{-4Q}{8\pi\epsilon_0 R}\right) + \frac{9Q}{12\pi\epsilon_0 R} = \frac{Q}{2\pi\epsilon_0 R} = 3.60 \times 10^8 \text{ V}$$

Q5 12 points

a)

$$Q_1 = C_1 V = \frac{\kappa\epsilon_0 A}{d} V = \frac{(3.00) \times (8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2) \times (5.00 \times 10^{-3} \text{ m}^2)}{2.00 \text{ mm}} (10.0 \text{ V}) = 6.64 \times 10^{-10} \text{ C}$$

$$Q_2 = C_2 V = \frac{\epsilon_0 A}{d} V = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2) \times (5.00 \times 10^{-3} \text{ m}^2)}{2.00 \text{ mm}} (10.0 \text{ V}) = 2.21 \times 10^{-10} \text{ C}$$

b)

$$E = \frac{V}{d} = \frac{10.0 \text{ V}}{2.00 \text{ mm}} = 5000 \text{ V/m or N/C}$$

$$\text{or } E = \frac{Q_1}{A\epsilon_0} = 5000 \text{ V/m or N/C}$$

c)

$$q' = Q_2 - Q_1 = -4.43 \times 10^{-10} \text{ C}$$

$$\text{or } q' = -\left(1 - \frac{1}{\kappa}\right) Q_2 = -4.43 \times 10^{-10} \text{ C}$$

Q6 10 points

$$\text{For left wire: two semi-infinite } B_1 = 2 \times \frac{1}{2} \frac{\mu_0 i}{2\pi R} = \frac{\mu_0 i}{2\pi R}, \quad \text{arc } B_2 = \frac{\mu_0 i \frac{\pi}{2}}{4\pi R} = \frac{\mu_0 i}{8R}$$

$$\text{For right wire: two semi-infinite } B_3 = 0, \quad \text{arc } B_4 = -\frac{\mu_0 i \frac{\pi}{2}}{4\pi R} = -\frac{\mu_0 i}{8R}$$

$$\text{So } B_{\text{net}} = \frac{\mu_0 i}{2\pi R} + \frac{\mu_0 i}{8R} - \frac{\mu_0 i}{8R} = \frac{\mu_0 i}{2\pi R}$$

Q7 12 pointsa) $E = 0$

b) Use Gauss' law

$$E \cdot 4\pi r^2 = \frac{q_1 + \int_b^r \frac{A}{r} \cdot 4\pi r^2 dr}{\epsilon_0}$$

$$E = \frac{q_1}{4\pi r^2} + \frac{A}{2} - \frac{Ab^2}{2r^2}$$

$$\Rightarrow A = \frac{q_1}{2\pi b^2} = 4.48 \times 10^{-12} \text{ C/m}^2$$

Q8 10 points

a) The rotating cylinder is just like a solenoid, so we can calculate the magnetic field inside it based on $B = \mu_0 n i$.

For the cylinder, the amount of charge that passes a given segment of length l on the cylinder in a time dt is $dq = \sigma l(\omega R) dt$. So the current per unit length is

$$J_s = \frac{1}{l} \frac{dq}{dt} = \sigma \omega R$$

So the field inside the cylinder is $B = \mu_0 \sigma \omega R$

b)

Consider a circular strip with width dx , a slant-distance x from the tip. The velocity of any point in this strip is $v = \omega x \sin \theta$, the current in the strip is

$$i = \frac{dq}{dt} = \frac{\sigma dx (v dt)}{dt} = \sigma \omega x \sin \theta dx$$

The magnetic field at the tip due to the strip is

$$dB = \frac{\mu_0 (2\pi x \sin \theta)}{4\pi x^2} \sin \theta = \frac{1}{2} \mu_0 \sigma \omega \sin^3 \theta dx$$

So the net magnetic field at the tip due to the cone is $\int_0^L dB = \frac{1}{2} \mu_0 \sigma \omega \sin^3 \theta L$